

ibm370_processor_sop_v1.json

```
{
  "id": "SOP-IBM370-001",
  "title": "IBM System/370 Model 145 Processor Check & Hard Stop Diagnostic",
  "system": "IBM System/370 Model 145",
  "component": "3145 Processing Unit",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Field Engineering Maintenance Manual (SY24-3581)",
  "description": "Diagnostic procedure for isolating internal processor checks and unplanned hard
stops on the Model 145.",
  "steps": [
    {
      "step": 1,
      "action": "Machine Check Analysis",
      "details": "When a 'Processor Check' occurs, examine the Machine Check Interruption
Code (MCIC) stored in real memory locations 232-239.",
      "indicators": {
        "Byte 0, Bit 1": "System Damage (Hard Failure)",
        "Byte 0, Bit 2": "Instruction Processing Damage",
        "Byte 1, Bit 1": "External Damage (Channels/Memory)"
      }
    },
    {
      "step": 2,
      "action": "Console Register Display",
      "details": "Use the console to display the Micro-Instruction Address Register (MIAR).
This indicates the specific micro-routine executing at the time of failure.",
      "check": "Record the 16-bit MIAR address for cross-reference with the Logic Diagrams
(ALD).",
    },
    {
      "step": 3,
      "action": "Core/WCS Parity Check",
      "details": "The Model 145 uses Writable Control Storage (WCS). Verify parity indicators
on the console for WCS locations.",
      "logic": "If WCS parity is tripped, reload the microprogram from the console file
(floppy disk).",
    },
    {
      "step": 4,
      "action": "FLT (Fault Location Test) Execution",
      "details": "Load the FLT microprograms from the system console file. Run the tests in
'Stop on Error' mode.",
      "remedy": "The FLT output will point to a specific SLT (Solid Logic Technology) card or
MST (Monolithic System Technology) logic board for replacement."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "MIAR stops at 0000",
      "next": "Internal micro-reset failure. Check console file loading."
    }
  ]
}
```

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    {
      "condition": "WCS Parity error",
      "next": "Reload microprogram. If recurs, replace WCS storage card."
    },
    {
      "condition": "Instruction Damage bit ON",
      "next": "Run CPU Logic FLTs to isolate failing MST module."
    }
  ]
}
```